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Low earth orbit satellite communication using device diversity reception and transmission

Abstract

A system for low earth orbit (LEO) satellite communication. In one example, the system includes a control computer and a workgroup including a plurality of communication devices. Each communication device includes an electronic processor configured to establish a wireless network connection with the control computer, determine connectivity data associated with a signal between the communication device and a first LEO satellite, and transmit the connectivity data to the control computer via the wireless network connection. The control computer is configured to receive the connectivity data from each communication device, and select a first one of the plurality of communication devices as a primary communication device based on each of the connectivity data. The primary communication device is configured to receive media from a different one of the plurality of communication devices via the wireless network connection and transmit the media to the first LEO satellite.

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Background/Summary

BACKGROUND

(1) Low earth orbit (LEO) satellite communication systems, otherwise referred to as device-to-satellite communication systems, enable terrestrial wireless communication devices to operate in off-terrestrial network scenarios (e.g., in rural areas without reliable terrestrial cellular networks). Reliability of LEO satellite communication is constrained by link budgets. Link budgets are an accounting of all power gains and losses that a communication signal experiences in a telecommunication system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments, examples, aspects, and features that include the claimed subject matter, and explain various principles and advantages.
- (2) FIG. 1 illustrates a system for low earth orbit (LEO) satellite communication, according to some aspects.
- (3) FIG. 2 illustrates a communication device included in the system of FIG. 1, according to some aspects.
- (4) FIG. 3 illustrates a workgroup control computer included in the system of FIG. 1, according to some aspects.
- (5) FIG. 4 illustrates a system for low earth orbit (LEO) satellite communication, according to some aspects.
- (6) FIG. 5 illustrates a first method for performing a LEO satellite communication, according to some aspects.

(7) FIG. 6 illustrates a second method for performing LEO satellite communication, according to some aspects.

(8) FIG. 7 illustrates a primary communication device selection method, according to some aspects.

(9) FIG. 8 illustrates a LEO satellite communication environment, according to some aspects.

(10) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments, examples, aspects, and features.

(11) The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments, examples, aspects, and features described and illustrated so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

(12) Link margin is the amount by which the received power of a signal exceeds the sensitivity of a receiver (e.g., a satellite). However, if the link margin of a communication link is sufficiently large, the communication link will be useful for sending data. Link budgets in many low earth orbit (LEO) satellite communication systems are typically constrained such that satellite-to-device communications are limited to minimally-occurring, non-realtime text messages. For example, terrestrial communication devices communicating with LEO satellites typically must be in open areas (e.g., open fields) without line-of-sight (LoS) obstructions between the communication device and the satellite in order to achieve a sufficient link margin. Additionally, a LEO satellite is typically only within range of a particular communication device for approximately 3-5 minutes at a time. In some instances, a user may be required to point the communication device in a particular direction (e.g., towards the satellite) in order to achieve a sufficient link margin for sending data to the satellite.

(13) To address, among other things, these problems, systems and methods are provided herein for device diversity reception and transmission in a device-to-satellite communication system. Among other things, examples described herein provide a system for low earth orbit (LEO) satellite communication. The system includes a control computer and a workgroup including a plurality of communication devices. Each respective communication device of the plurality of communication devices includes an electronic processor configured to establish a wireless network connection with the control computer, determine connectivity data associated with a signal between the respective communication device and a first LEO satellite, and transmit the connectivity data to the control computer via the wireless network connection. The control computer is configured to receive the connectivity data from each communication device of the plurality of communication devices, and select a first one of the plurality of communication devices as a primary communication device based on each of the connectivity data. The primary communication device is configured to establish a communication link with the first LEO satellite, receive media from a different one of the plurality of communication devices via the wireless network connection, and transmitting the media to the first LEO satellite.

(14) In some aspects, the control computer is included in at least one of the plurality of communication devices.

(15) In some aspects, selecting the first one of the plurality of communication devices as the primary communication device includes determining, based on the connectivity data, that the first one of the plurality of communication devices is in a field of view of the first LEO satellite.

(16) In some aspects, the control computer is further configured to periodically receive connectivity data updates from each communication device of the plurality of communication devices, store each of the connectivity data updates, predict future connectivity data for each communication device of the plurality of communication devices based on the connectivity data updates, and select

a second one of the plurality of communication devices as an upcoming primary communication device based on the future connectivity data.

(17) In some aspects, the future connectivity data includes a predicted visibility time of a second LEO satellite to the upcoming primary communication device, and the control computer is further configured to transmit a command to the upcoming primary communication device to establish a communication link with the second LEO satellite.

(18) In some aspects, the connectivity data includes at least one selected from the group consisting of a signal strength of the signal between the respective communication device and the first LEO satellite, a quality of service (QoS) of the signal between the respective communication device and the first LEO satellite, an estimated remaining connection time with the first LEO satellite, and obstruction data.

(19) In some aspects, the control computer is further configured to select a second one of the plurality of communication devices as a secondary communication device based on the connectivity data, periodically receive connectivity data updates from the plurality of communication devices, determine, based on the connectivity data updates, whether the QoS between the primary communication device and the first LEO satellite meets a QoS threshold, in response to determining, based on the connectivity data, that the QoS between the primary communication device and the first LEO satellite is below the QoS threshold, select the secondary communication device as the primary communication device.

(20) In some aspects, selecting the secondary communication device as the primary communication device includes transmitting a command to the secondary communication device to establish a communication link with a second LEO satellite, wherein the secondary communication device is in a field of view of the second LEO satellite.

(21) In some aspects, the wireless network connection includes at least one selected from the group consisting of a Wi-Fi connection, a Bluetooth connection, an 802.15.4g connection, a land mobile radio (LMR) connection, an ultra-wideband (UWB) connection, and a long term evolution (LTE) direct connection.

(22) In some aspects, the plurality of communication devices includes a camera-equipped communication device, the camera-equipped communication device includes a camera configured to capture image data of an environment surrounding the camera-equipped communication device, and the control computer is configured to receive, from the camera-equipped communication device, sensor data based on the image data, and determine a satellite line-of-sight (LoS) obstruction based on the sensor data.

(23) In some aspects, the satellite line-of-sight obstruction includes a predicted satellite LoS obstruction, and the control computer is configured to predict a path of the first LEO satellite and determine the predicted satellite LoS obstruction based on a predicted path of the first LEO satellite.

(24) In some aspects, the control computer is configured to in response to determining the satellite LoS obstruction, determine, based on the image data and a predicted path of the first LEO satellite, an unobstructed location, and transmit a message to the primary communication device, the message including instructions for a user of the primary communication device to move to the unobstructed location.

(25) In some aspects, the control computer is configured to predict a path of a second LEO satellite, and in response to determining the satellite LoS obstruction, determine, transmit a command to the primary communication device to establish a communication link with the second LEO satellite.

(26) In some aspects, the control computer is configured to in response to determining the satellite LoS obstruction, select a second one of the plurality of communication devices as an upcoming primary communication device.

(27) Another example provides, a method for low earth orbit (LEO) satellite communication. The method includes establishing, with each respective communication device of a plurality of

communication device, a wireless network connection with a control computer, determining, with each respective communication device of a plurality of communication devices, connectivity data associated with a signal between the respective communication device and a first LEO satellite, and transmitting, with each respective communication device of the plurality of communication devices, the connectivity data to the control computer via the wireless network connection. The method also includes receiving, with the control computer, the connectivity data from each communication device of the plurality of communication devices, and selecting, with the control computer, a first one of the plurality of communication devices as a primary communication device based on each of the connectivity data. The method further includes establishing, with the primary communication device, a communication link with the first LEO satellite, receiving, with the primary communication device, media from a different one of the plurality of communication devices via the wireless network connection, and transmitting, with the primary communication device, the media to the first LEO satellite.

(28) In some aspects, the control computer is included in at least one of the plurality of communication devices.

(29) In some aspects, selecting the first one of the plurality of communication devices as the primary communication device includes determining, based on the connectivity data, that the first one of the plurality of communication devices is in a field of view of the first LEO satellite.

(30) In some aspects, the method further includes periodically receiving, with the control computer, connectivity data updates from each communication device of the plurality of communication devices, storing, with the control computer, each of the connectivity data updates, predicting with the control computer, future connectivity data for each communication device of the plurality of communication devices based on the connectivity data updates, and selecting with the control computer, a second one of the plurality of communication devices as an upcoming primary communication device based on the future connectivity data.

(31) In some aspects, the future connectivity data includes a predicted visibility time of a second LEO satellite to the upcoming primary communication device, and the method further comprises transmitting, with the control computer, a command to the upcoming primary communication device to establish a communication link with the second LEO satellite.

(32) In some aspects, the connectivity data includes at least one selected from the group consisting of a signal strength of the signal between the respective communication device and the first LEO satellite, a quality of service (QoS) of the signal between the respective communication device and the first LEO satellite, an estimated remaining connection time with the first LEO satellite, and obstruction data.

(33) For ease of description, some or all of the example systems presented herein are illustrated with a single exemplar of each of its component parts. Some examples may not describe or illustrate all components of the systems. Other examples may include more or fewer of each of the illustrated components, may combine some components, or may include additional or alternative components.

(34) It should be understood that although certain figures presented herein illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. In some examples, the illustrated components may be combined or divided into separate software, firmware, and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided as shown and unless the context clearly indicates otherwise, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links.

(35) FIG. 1 illustrates an example system **10** for LEO satellite communication, according to some aspects. The system **10** may be implemented in an environment out of range of radio or cell towers

or may be implemented in an environment where cellular network signals and/or radio signals are unreliable. The system **10** includes one or more LEO satellites **12**, for example, LEO satellites **12.sub.1, 2, . . . , n**, configured to orbit around the earth with a period of, for example, 128 minutes or less. The system **10** illustrated includes three LEO satellites but may include a different number of LEO satellites. For example, the system **10** may include a constellation of **10** satellites, **30** satellites, **60** satellites, or another suitable number of satellites. Some or all of the LEO satellites **12** may be in communication with each other via an inter-satellite link (ISL). In some instances, some or all of the LEO satellites **12** also communicate with a cloud-based server **14**.

(36) In the example shown, the system **10** also includes a communication group, or workgroup **18**, having a plurality of communication devices **20** and a workgroup control computer **22**. The plurality of communication devices **20** included in the workgroup **18** may be, for example, communication devices used by first responders (for example, fire fighters, police officers, and paramedics, among others) for communicating with one another and with a dispatcher. The plurality of communication devices **20** may include one or more radios (e.g., land mobile radios (LMRs)), one or more mobile phones, and/or other communication devices capable of wireless communication. In the illustrated example, the plurality of communication devices **20** includes three communication devices, however, the plurality of communication devices **20** may include more than three communication devices or less than three communication devices. In some instances, the number of communication devices included in the workgroup **18** is dependent on the separation between each of the communication devices with one another. For example, each of the plurality of communication devices **20** may be within a range of one another such that each of the plurality of communication devices **20** is operable to join a wireless network **24** via wireless network connection.

(37) The wireless network **24** may be a short-range wireless network enabling each of the plurality of communication devices **20** to communicate with one another in the absence of a cellular network. The wireless network **24** may include, for example, a Wi-Fi network, a Bluetooth network, an 802.15.4g network, a land mobile radio (LMR) network, an ultra-wideband (UWB) network, a long term evolution (LTE) direct network (e.g., LTE proximity services (ProSe) network), and/or another network. Thus, the wireless network connection may include, for example, a Wi-Fi connection, a Bluetooth connection, an 802.15.4g connection, a land mobile radio (LMR) connection, an ultra-wideband (UWB) connection, a LTE direct network (e.g., a LTE ProSe connection), and/or other wireless network connections. Some or all of the communication devices included in the plurality of communication devices **20** may be further configured to establish a satellite communication link **28** with one or more of the LEO satellites **12**.

(38) FIG. 2 schematically illustrates an example communication device **20a** that may be included in the plurality of communication devices **20**, according to some aspects. In the example illustrated, the communication device **20a** includes an electronic processor **38** (for example, a microprocessor or other electronic device). The electronic processor **38** is electrically coupled to a memory **42**, a communication interface **48**, and a user interface **52**. The communication device **20a** may include fewer or additional components in configurations different from that illustrated in FIG. 2. For example, the communication device **20a** may additionally include a push-to-talk button. The user interface **52** may include, for example, a camera, a push-to-talk button, a keyboard, a keypad, one or more speakers, a microphone, a display, and/or other input and output devices. In some examples, the communication device **20a** includes a location component (for example, a global positioning system receiver) configured to determine the geographic coordinates of the location of the communication device **20a**. In some examples, the communication device **20a** performs functionality other than the functionality described below.

(39) The memory **42** may include read only memory (ROM), random access memory (RAM), other non-transitory computer-readable media, or a combination thereof. In the example shown, the memory **42** stores a connection algorithm **44** executable by the electronic processor **38** for

performing methods described herein.

(40) The communication interface **48** sends and receives data to and from (for example, via an antenna) other devices in the system **10** over the wireless network **24**. The communication interface **48** is also operable to send and receive data to and from the LEO satellites **12** over the satellite communication link **28**. The electronic processor **38** is configured to transmit and receive (for example, via the communication interface **48** and the antenna) media (for example, a video stream, audio stream, an image, a text message, sensor input data gathered by one or more sensors of the communication device **20a**, and the like) to and from a different one of the plurality of communication devices **20**, the workgroup control computer **22**, and/or a LEO satellite **12**. Similarly, the electronic processor **38** may output media received from a different one of the plurality of communication devices **20**, the workgroup control computer **22**, and/or a LEO satellite **12** through the user interface **52** or a combination thereof.

(41) The electronic processor **38** transmits connectivity data to the workgroup control computer **22**. The connectivity data includes, for example, a signal strength of a signal between the communication device **20a** and a LEO satellite **12**, a quality of service (QoS) of a signal between the communication device **20a** and the LEO satellite **12**, a duration of connection time with the LEO satellite **12**, an estimated remaining connection time with the LEO satellite **12**, information related to a LoS obstruction with the LEO satellite **12**, and/or other information associated with connectivity to the LEO satellite **12**. In some instances, some or all of the connectivity data is included in the media transmitted from the communication device **20a** to the workgroup control computer **22**. In some instances, the workgroup control computer **22** extrapolates the connectivity data based on a quality of the media, a duration of the media, metadata included in the media, and/or the contents of the media.

(42) FIG. 3 schematically illustrates an example of the workgroup control computer **22**, according to some aspects. The workgroup control computer **22** includes, for example, electronic processor **64** electrically coupled to a memory **68** and a communication interface **72**. The memory **68** may include read only memory (ROM), random access memory (RAM), other non-transitory computer-readable media, or a combination thereof. The communication interface **72** is configured to transmit and receive media and connectivity data from the plurality of communication devices **20** over the wireless network **24**. In some instances, the communication interface **72** is also configured to transmit and receive media to and from a LEO satellite **12** over the satellite communication link **28**. The memory **68** is configured to store the connection algorithm **44** executable by the electronic processor **64** for performing methods described herein. The memory **68** is also configured to store historical connectivity data **76** associated with currently and previously received connectivity data from the plurality of communication devices **20**. The workgroup control computer **22** is also configured to predict the path of one or more of the LEO satellites **12**, based on, for example, historical connectivity data **76** associated with previous passes of the one or more LEO satellites **12** over the workgroup **18**. The workgroup control computer **22** may also include a user interface **80**.

(43) The workgroup control computer **22** may include more or fewer components than those illustrated in FIG. 3. Although illustrated as a separate device in FIG. 1, the workgroup control computer **22** may be included in at least one of the plurality of communication devices **20**. In one example, the workgroup control computer **22** is implemented in a distributed manner in some or all of the plurality of communication devices **20**. In another example, the workgroup control computer **22** is implemented in a single one of the plurality of communication devices **20**. For example, the workgroup control computer **22** may be implemented in a communication device having the largest battery capacity, having the largest processing capacity, and/or having access to a power source. For example, a communication device implemented in a vehicle may have a longer battery life than, for example, a hand-held radio and the control computer is implemented on that device, for example, by installing appropriate software on that device. In some instances, the plurality of communication devices **20** collectively designate a particular one of the plurality of communication

devices **20** as the workgroup control computer **22**. In some instances, the workgroup control computer **22** is manually selected, for example, by a user of the workgroup **18**.

(44) Satellite communication may be affected by obstructions between a satellite and a terrestrial communication device. Additionally, because the location of LEO satellites changes over time, maintaining a satellite communication link over longer periods of time is typically not possible. Therefore, the workgroup control computer **22** determines which of the plurality of communication devices **20** has the strongest connection with a satellite based on connectivity data received from each of the plurality of communication devices **20**. Rather than each of the plurality of communication devices **20** attempting to establish a satellite communication link **28** with a satellite as needed, the workgroup control computer **22** determines which of the plurality of communication devices **20** has a strongest connection with a LEO satellite.

(45) Referring now to FIG. **4**, the workgroup control computer **22** designates a first one of the communication devices **20** having the strongest connection to a satellite, for example, a first LEO satellite **12.sub.1** illustrated in FIG. **1**, as a primary communication device **20.sub.1**. The primary communication device **20.sub.1** is responsible for transmitting and receiving, over the satellite communication link **28**, all media for the workgroup **18**.

(46) The workgroup control computer **22** may additionally select a secondary communication device **20.sub.2**. The secondary communication device **20.sub.2** may have a second strongest connection to a satellite (e.g., either of the first LEO satellite **12.sub.1** or a second LEO satellite **12.sub.2** illustrated in FIG. **1**). The secondary communication device **20.sub.2** is configured to take over the role of the primary communication device **20.sub.1** in the event of poor transmission or reception quality from the primary communication device **20.sub.1**. The workgroup control computer **22** may further select a tertiary communication device **20.sub.3**. The tertiary communication device **20.sub.3** may have a third strongest connection to the first LEO satellite **12.sub.1**, the second LEO satellite **12.sub.2**, or another LEO satellite **12.sub.n**. The tertiary communication device **20.sub.3** is configured to take over the role of the primary communication device **20.sub.1** in the event of poor transmission or reception quality from both the primary communication device **20.sub.1** and the secondary communication device **20.sub.2**. In some instances, the workgroup control computer **22** stores (e.g., in the memory **68**) a tiered device list ranking each of the plurality of communication devices **20** (e.g., communication devices **20.sub.1-20.sub.n**) based on the connectivity data.

(47) FIG. **5** illustrates a method **100** performed by the primary communication device **20.sub.1** for transmitting media to a LEO satellite (e.g., the first LEO satellite **12.sub.1**), according to some examples. At block **104**, the primary communication device **20.sub.1** receives media from a different one of the plurality of communication devices **20** (e.g., the secondary communication device **20.sub.2**) via the wireless communication network **24**. The media is generated by the different one of the plurality of communication devices **20** and intended to be transmitted to one or more devices external to the system **10**. The media stream is, for example, an audio stream or a video stream sent to a call dispatch center. In some instances, the primary communication device **20.sub.1** may receive the media directly from the different one of the plurality of communication devices **20**. In some instances, the workgroup control computer **22** receives the media from the different one of the plurality of communication devices **20** over the wireless communication network **24** and sends the media to the primary communication device **20.sub.1** for transmission to the first LEO satellite **12.sub.1**. In some instances, the media is generated by the primary communication device **20.sub.1**. At block **108**, after receiving the media, the primary communication device **20.sub.1** transmits a message (including the media) to the first LEO satellite **12.sub.1** with which the primary communication device **20.sub.1** has established a satellite communication link **28**.

(48) FIG. **6** illustrates a method **200** performed by the primary communication device **20.sub.1** for receiving media from a LEO satellite (e.g., the first LEO satellite **12.sub.1**), according to some

examples. At block **204** the primary communication device **20.sub.1** receives a message from the first LEO satellite **12.sub.1**. The message includes media intended to be received by a particular one of the plurality of communication devices **20**. At block **208**, the primary communication device **20.sub.1** sends the media to the particular one of the plurality of communication devices **20** via the wireless network **24**. In some instances, the primary communication device **20.sub.1** transmits the media directly to the particular one of the plurality of communication devices **20**. In some instances, the primary communication device **20.sub.1** transmits the media to workgroup control computer **22**, and the workgroup control computer **22** transmits the media to the particular one of the plurality of communication devices **20**.

(49) By selecting a particular communication device that is operable to communicate with a satellite **12** as the primary communication device **20.sub.1**, others of the plurality of communication devices **20** are able to transmit and receive messages with the satellite **12** (e.g., via the primary communication device **20.sub.1**) without necessarily having satellite communication capability. For example, a device not capable of performing satellite transmission may instead transmit a message to the primary communication device **20.sub.1**, which the primary communication device **20.sub.1** then forwards to the first LEO satellite **12.sub.1**.

(50) FIG. **7** illustrates a method **300** performed by the workgroup **18** for selecting a primary communication device **20.sub.1**, according to some examples. Although the method **300** is described in conjunction with the system **10** as described herein, the method **300** may be used with other systems and devices. For example, while the method **300** is described with respect to the plurality of communication devices **20** and the workgroup control computer **22**, it should be understood that the workgroup control computer **22** may be implemented in at least one of the plurality of communication devices **20**. In addition, the method **300** may be modified or performed differently than the specific example provided. For example, the order of the method described herein may be changed, and one or more method paths may be implemented in parallel with one another.

(51) At block **304**, the plurality of communication devices **20** and the workgroup control computer **22** establish the wireless network connection via the wireless network **24**. At block **308**, each respective communication device of the plurality of communication devices **20** determines initial connectivity data associated with a signal between the respective communication device and at least one LEO satellite **12**. The LEO satellite **12** may be a satellite having a field of view including the communication device. As described above, the connectivity data may include a signal strength value associated with a signal between a respective communication device and the LEO satellite **12**, a quality of service associated with the signal, a remaining connection time with the LEO satellite **12**, obstruction data associated with an obstruction of the signal, or other connection metrics.

(52) At block **312**, each of the plurality of communication devices **20** transmits the connectivity data to the workgroup control computer **22** over the wireless network **24**. At block **316**, based on the connectivity data received from each of the plurality of communication devices **20**, the workgroup control computer **22** selects a first one of the plurality of communication devices **20** as the primary communication device **20.sub.1**. The workgroup control computer **22** may also select a second one of the plurality of communication devices **20** as the secondary communication device **20.sub.2** and select a third one of the plurality of communication devices **20** as the tertiary communication device **20.sub.3**. In some instances, the workgroup control computer **22** generates, based on the connectivity data, a tiered device list ranking the connection strength of each of the plurality of communication devices **20**.

(53) After selecting the primary communication device **20.sub.1**, the secondary communication device **20.sub.2**, and the tertiary communication device **20.sub.3**, the workgroup control computer **22** initiates a predictive loop **318**. During the predictive loop **318**, the workgroup control computer **22** periodically requests connectivity data updates from each of the plurality of communication

devices **20** (in block **320**). At block **324**, the workgroup control computer **22** stores the connectivity data updates in the memory **68**, for example, as the historical connectivity data **76**. The workgroup control computer **22** may also store the first connectivity data in the memory **68** as the historical connectivity data **76**. Based on the historical connectivity data **76**, the workgroup control computer **22** predicts future connectivity data associated with some or all of the plurality of communication devices **20** (at block **328**). The workgroup control computer **22** may include a learning module for predicting the paths of some or all of LEO satellites **12**, for example, the first LEO satellite **12.sub.1** and the second LEO satellite **12.sub.2**, based on the historical connectivity data **76**.

Alternatively, or in addition, the workgroup control computer **22** may store a data file including, for example, ephemeris data associated with some or all of the LEO satellites **12**. For each respective one of the plurality of communication devices **20**, the workgroup control computer **22** may predict a visibility time remaining of a LEO satellite (e.g., the first LEO satellite **12.sub.1**), and predict an upcoming visibility of another LEO satellite (e.g., the second LEO satellite **12.sub.2**).

(54) Based on the predicted future connectivity data, the workgroup control computer **22** determines an upcoming primary communication device (at block **332**). For example, the workgroup control computer **22** predicts which of the plurality of communication devices **20** will have the strongest connection to a LEO satellite **12** after, for example, the first LEO satellite **12.sub.1** moves out of the field of view of the primary communication device **20.sub.1**. In some instances, the upcoming primary communication device is a different one of the plurality of communication devices **20** than the primary communication device **20.sub.1**. Based on the future connectivity data, the workgroup control computer **22** may predict that the upcoming primary communication device will have the strongest connection to the first LEO satellite **12.sub.1**. In some instances, the workgroup control computer **22** predicts, based on the future connectivity data, that the upcoming primary communication device will have the strongest connection with the second LEO satellite **12.sub.2**. In some instances, the workgroup control computer **22** predicts that the primary communication device **20.sub.1** is the upcoming primary communication device, but that the primary communication device **20.sub.1** will have the strongest connection with the second LEO satellite **12.sub.2** rather than the first LEO satellite **12.sub.1**. Similarly, the workgroup control computer **22** may also determine an upcoming secondary communication device and an upcoming tertiary communication device based on the future connectivity data.

(55) At block **338**, the workgroup control computer **22** hands off designation of the primary communication device **20.sub.1** to the upcoming primary communication device, designation of the secondary communication device **20.sub.2** to the upcoming secondary communication device, and designation of the tertiary communication device **20.sub.3** to the upcoming tertiary communication device. The workgroup control computer **22** may reorder the tiered device list before the satellite communication link **28** between the primary communication device **20.sub.1** and the first LEO satellite **12.sub.1** is lost. Therefore, the risk of communication interruption in the workgroup **18** is reduced. In some instances, the workgroup control computer **22** determines a specific time to perform the device handoff. After reordering the tiered device list (in block **338**), the predictive loop **318** repeats.

(56) After selecting the primary communication device **20.sub.1**, the secondary communication device **20.sub.2**, and the tertiary communication device **20.sub.3** (at block **316**), the workgroup control computer **22** also initiates a watchdog loop **340** for monitoring connection quality of the primary communication device **20.sub.1** to the LEO satellite **12** having the strongest connection to the primary communication device **20.sub.1**, for example, the first LEO satellite **12.sub.1**. During the watchdog loop **340**, the primary communication device **20.sub.1** establishes a satellite communication link **28** with the first LEO satellite **12.sub.1**, and, at block **344**, performs LEO satellite communication with the first LEO satellite **12.sub.1** on behalf of the workgroup **18**.

(57) During and/or after performing the LEO satellite communication, for example using the method **100** described with respect to FIG. **5** and/or the method **200** described with respect to FIG.

6, the workgroup control computer **22** determines whether the QoS associated with the communication link **28** meets or exceeds a QoS threshold. For example, during the period of time since the selection of the primary communication device **20.sub.1**, the location of the primary communication device **20.sub.1** relative to the first LEO satellite **12.sub.1** may have changed, or an obstruction may have impacted the LoS between the primary communication device **20.sub.1** and the first LEO satellite **12.sub.1**. Therefore, the workgroup control computer **22** periodically receives connectivity data associated with the satellite communication link **28** between the primary communication device **20.sub.1** and the first LEO satellite **12.sub.1**.

(58) When the workgroup control computer **22** determines that QoS associated with the communication performed by the primary communication device **20.sub.1** meets or exceeds the QoS threshold (in block **348**), the workgroup control computer **22** determines whether the wireless network connectivity of the primary communication device **20.sub.1** to others of the plurality of communication devices **20** over the wireless network **24** meets or exceeds a wireless network connectivity threshold (in block **352**). For example, the wireless network connection between the primary communication device **20.sub.1** and the others of the plurality of communication devices **20** may not be sufficiently strong to enable the primary communication device **20.sub.1** to transmit and receive media to and from the others of the plurality of communication devices **20**.

(59) When the workgroup control computer **22** determines that the wireless network connectivity of the primary communication device **20.sub.1** is below the wireless network connectivity threshold (in block **352**), the workgroup control computer **22** designates the secondary communication device **20.sub.2** as the primary communication device **20.sub.1** (in block **356**). In some instances, the workgroup control computer **22** may reorder the tiered device list ranking based on connectivity data periodically received from each of the plurality of communication devices **20**. For example, the workgroup control computer **22** may designate the secondary communication device **20.sub.2** as a primary communication device, designate the tertiary communication device **20.sub.3** as a secondary communication device, and designate the primary communication device **20.sub.1** as a tertiary communication device. Similarly, when the workgroup control computer **22** determines in block **348** that the QoS of the satellite communication link **28** between the initially assigned primary communication device **20.sub.1** and the first LEO satellite **12.sub.1** does not meet or exceed the QoS threshold (in block **348**), the workgroup control computer **22** designates the secondary communication device **20.sub.2** as the primary communication device **20.sub.1** (in block **352**). After the workgroup control computer **22** assigns the secondary communication device **20.sub.2** as a primary communication device **20.sub.1** (in block **352**), the watchdog loop **340** repeats and the new primary communication device **20.sub.1** performs LEO satellite communication (in block **344**).

(60) When the workgroup control computer **22** determines that the QoS associated with the satellite communication link **28** meets or exceeds the QoS threshold (in block **348**) and that the network connectivity meets or exceeds the network connectivity threshold (in block **352**), the watchdog loop **340** does not proceed to block **356**. Instead, the watchdog loop **340** proceeds to block **344**, and the initially assigned primary communication device **20.sub.1** continues to perform LEO satellite communication on behalf of the workgroup **18**.

(61) In the example provided, the method **300** additionally includes a sensor loop **364**. During the sensor loop **364**, the workgroup control computer **22** receives sensor data from some or all of the plurality of communication devices **20** (at block **368**). The sensor data may include image data, such as for example a video stream, generated by the camera of one or more of the plurality of communication devices **20**. The workgroup control computer **22** may perform video analytics on the sensor data and determine whether an obstruction is detected in the LoS of a particular communication device (at block **372**). For example, based on the sensor data, the workgroup control computer **22** may determine that the environment surrounding the primary communication device **20.sub.1** includes an obstruction (e.g., foliage, buildings, etc.) in the LoS of the primary

communication device **20.sub.1** and the first LEO satellite **12.sub.1**. Alternatively, or in addition, each respective one of the plurality of communication devices **20** performs an obstruction analysis on respective sensor data, and transmits a result of the analysis to the workgroup control computer **22**. For example, a particular one of the plurality of communication devices **20** may determine that the particular communication device has a 20% obstruction with a satellite.

(62) The sensor data may be generated at the primary communication device **20.sub.1**, or at a different one of the plurality of communication devices **20** having a view of the primary communication device **20.sub.1**. In some instances, the workgroup control computer **22** predicts a satellite LoS obstruction based on, for example, the sensor data and the predicted path of the first LEO satellite **12.sub.1**. The workgroup control computer **22** may also determine an obstruction for the secondary communication device **20.sub.2** and the tertiary communication device **20.sub.3**. In some instances, the workgroup control computer **22** is configured to determine an obstruction for each of the plurality of communication devices **20**. In some instances, the workgroup control computer **22** includes a learning module for predicting an obstruction based on previously received sensor data.

(63) If the workgroup control computer **22** does not detect an obstruction or an upcoming obstruction, the sensor loop returns to block **368** and repeats. If, on the other hand, the workgroup control computer **22** does determine an obstruction or an upcoming obstruction in the LoS of, for example, the primary communication device **20.sub.1** and the first LEO satellite **12.sub.1**, the workgroup control computer **22** determines a control response based on the obstruction (at block **376**). In some instances, the control response includes performing a handoff of the primary communication device **20.sub.1**. For example, the workgroup control computer **22** may reorder the tiered device list and reassign the primary communication device **20.sub.1**, the secondary communication device **20.sub.2**, and/or the tertiary communication device **20.sub.3**. In some instances, such as when each of the plurality of communication devices **20** has a poor connection with the LEO satellites **12** or when only the primary communication device **20.sub.1** is otherwise available, the control response includes determining an unobstructed location within a vicinity (e.g., within 10 feet of the primary communication device **20.sub.1**, within 50 feet of the primary communication device **20.sub.1**, within a mile of the primary communication device **20.sub.1**, or another suitable distance from the primary communication device **20.sub.1**). The workgroup control computer **22** transmits a message to the primary communication device **20.sub.1** via the wireless network connection. In some examples, the message includes instructions for a user of the primary communication device **20.sub.1** to move to the unobstructed location. The instructions may include a suggested route for the user of the primary communication device **20.sub.1** to reach the unobstructed location.

(64) In some instances, the workgroup control computer **22** determines an upcoming primary communication device, an upcoming secondary communication device, and/or an upcoming tertiary communication device based on the determined obstruction. In some instances, the workgroup control computer **22** provides a command to the primary communication device **20.sub.1** to establish a satellite communication link **28** with the second LEO satellite **12.sub.2** rather than the first LEO satellite **12.sub.1**. Although described herein as a separate loop, the sensor loop **364** may alternatively be implemented as part of one or both of the predictive loop **318** and the watchdog loop **340**.

(65) FIG. **8** illustrates an example environment **400** in which the system **10** may be implemented. As described above, a workgroup control computer may be implemented in at least one of the plurality of communication devices **20**. In the illustrated example, each of the plurality of communication devices **20** are in the field of view of the first LEO satellite **12.sub.1**. The primary communication device **20.sub.1** experiences high signal strength from the first LEO satellite **12.sub.1** and may be operable to deliver 5 minutes of high QoS transmission and reception of media with the first LEO satellite **12.sub.1**. The secondary communication device **20.sub.2**

experiences a moderate signal strength from the first LEO satellite **12.sub.1** due to an occlusion from a first obstruction **404**. In the illustrated example, the secondary communication device **20.sub.2** may be operable to deliver 0.8 minutes of moderate QoS transmission and reception of media with the first LEO satellite **12.sub.1**. The tertiary communication device **20.sub.3** experiences a low signal strength from the first LEO satellite **12.sub.1** due to an occlusion from a second obstruction **408**. In the illustrated example, the tertiary communication device **20.sub.3** may be operable to deliver 3 minutes of low quality transmission and reception of media with the first LEO satellite **12.sub.1**. A fourth, or quaternary communication device **20.sub.4** may experience low signal strength from the first LEO satellite **12.sub.1** due to occlusion from a third obstruction **412**. In the illustrated example, the quaternary communication device **20.sub.4** may be operable to deliver 0.5 minutes of low quality transmission and reception of media with the first LEO satellite **12.sub.1**.

(66) In the foregoing specification, specific examples have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the claimed subject matter. For example, in some instances, the workgroup control computer **22** may be at least partially implemented in the cloud, such as cloud-based server **14**. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings.

(67) The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims.

(68) Moreover, in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a,” “has . . . a,” “includes . . . a,” or “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially,” “essentially,” “approximately,” “about,” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting example the term is defined to be within 10%, in another example within 5%, in another example within 1% and in another example within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not listed.

(69) It will be appreciated that some examples may be comprised of one or more generic or specialized processors (or “processing devices”) such as microprocessors, digital signal processors, customized processors and field programmable gate arrays (FPGAs) and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of the method and/or apparatus described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application specific integrated circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic. Of course, a combination of the two

approaches could be used.

(70) Moreover, an example can be implemented as a computer-readable storage medium having computer readable code stored thereon for programming a computer (e.g., comprising a processor) to perform a method as described and claimed herein. Examples of such computer-readable storage mediums include, but are not limited to, a hard disk, a CD-ROM, an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

(71) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A system for low earth orbit (LEO) satellite communication, the system comprising: a control computer; and a workgroup including a plurality of communication devices, the plurality of communication devices including a camera-equipped communication device including a camera configured to capture image data of an environment surrounding the camera-equipped communication device; wherein each respective communication device of the plurality of communication devices includes an electronic processor configured to: establish a wireless network connection with the control computer, determine connectivity data associated with a signal between the respective communication device and a first LEO satellite, and transmit the connectivity data to the control computer via the wireless network connection; wherein the control computer is configured to: receive the connectivity data from each communication device of the plurality of communication devices, and select a first one of the plurality of communication devices as a primary communication device based on each of the connectivity data, periodically receive connectivity data updates from each communication device of the plurality of communication devices, store each of the connectivity data updates, predict future connectivity data for each communication device of the plurality of communication devices based on the connectivity data updates, select a second one of the plurality of communication devices as an upcoming primary communication device based on the future connectivity data, and receive, from the camera-equipped communication device, sensor data based on the image data, and determine a satellite line-of-sight (LoS) obstruction based on the sensor data; wherein the primary communication device is configured to: establish a communication link with the first LEO satellite, receive media from a different one of the plurality of communication devices via the wireless network connection, and transmitting the media to the first LEO satellite.
2. The system of claim 1, wherein the control computer is included in at least one of the plurality of communication devices.
3. The system of claim 1, wherein selecting the first one of the plurality of communication devices as the primary communication device includes determining, based on the connectivity data, that the

first one of the plurality of communication devices is in a field of view of the first LEO satellite.

4. The system of claim 1, wherein the future connectivity data includes a predicted visibility time of a second LEO satellite to the upcoming primary communication device, and the control computer is further configured to transmit a command to the upcoming primary communication device to establish a communication link with the second LEO satellite.

5. The system of claim 1, wherein the connectivity data includes at least one selected from the group consisting of a signal strength of the signal between the respective communication device and the first LEO satellite, a quality of service (QoS) of the signal between the respective communication device and the first LEO satellite, an estimated remaining connection time with the first LEO satellite, and obstruction data.

6. The system of claim 5, wherein the control computer is further configured to select a second one of the plurality of communication devices as a secondary communication device based on the connectivity data, periodically receive connectivity data updates from the plurality of communication devices, determine, based on the connectivity data updates, whether the QoS between the primary communication device and the first LEO satellite meets a QoS threshold, in response to determining, based on the connectivity data, that the QoS between the primary communication device and the first LEO satellite is below the QoS threshold, select the secondary communication device as the primary communication device.

7. The system of claim 6, wherein selecting the secondary communication device as the primary communication device includes transmitting a command to the secondary communication device to establish a communication link with a second LEO satellite, wherein the secondary communication device is in a field of view of the second LEO satellite.

8. The system of claim 1, wherein the wireless network connection includes at least one selected from the group consisting of a Wi-Fi connection, a Bluetooth connection, an 802.15.4g connection, a land mobile radio (LMR) connection, an ultra-wideband (UWB) connection, and a long term evolution (LTE) direct connection.

9. The system of claim 1, wherein the satellite line-of-sight obstruction includes a predicted satellite LoS obstruction, and the control computer is configured to predict a path of the first LEO satellite and determine the predicted satellite LoS obstruction based on a predicted path of the first LEO satellite.

10. The system of claim 1, wherein the control computer is configured to in response to determining the satellite LoS obstruction, determine, based on the image data and a predicted path of the first LEO satellite, an unobstructed location, and transmit a message to the primary communication device, the message including instructions for a user of the primary communication device to move to the unobstructed location.

11. The system of claim 1, wherein the control computer is configured to predict a path of a second LEO satellite, and in response to determining the satellite LoS obstruction, transmit a command to the primary communication device to establish a communication link with the second LEO satellite.

12. The system of claim 1, wherein the control computer is configured to in response to determining the satellite LoS obstruction, select a second one of the plurality of communication devices as an upcoming primary communication device.

13. A method for low earth orbit (LEO) satellite communication, the method comprising: establishing, with each respective communication device of a plurality of communication devices, a wireless network connection with a control computer; determining, with each respective communication device of a plurality of communication devices, connectivity data associated with a signal between the respective communication device and a first LEO satellite; transmitting, with each respective communication device of the plurality of communication devices, the connectivity data to the control computer via the wireless network connection; receiving, with the control computer, the connectivity data from each communication device of the plurality of communication

devices; selecting, with the control computer, a first one of the plurality of communication devices as a primary communication device based on each of the connectivity data; periodically receiving, with the control computer, connectivity data updates from each communication device of the plurality of communication devices, storing, with the control computer, each of the connectivity data updates, predicting with the control computer, future connectivity data for each communication device of the plurality of communication devices based on the connectivity data updates, and selecting with the control computer, a second one of the plurality of communication devices as an upcoming primary communication device based on the future connectivity data; receiving, from a camera-equipped communication device of the plurality of communication devices, sensor data based on the image data, and determine a satellite line-of-sight (LoS) obstruction based on the sensor data; establishing, with the primary communication device, a communication link with the first LEO satellite; receiving, with the primary communication device, media from a different one of the plurality of communication devices via the wireless network connection; and transmitting, with the primary communication device, the media to the first LEO satellite.

14. The method of claim 13, wherein the control computer is included in at least one of the plurality of communication devices.

15. The method of claim 13, wherein selecting the first one of the plurality of communication devices as the primary communication device includes determining, based on the connectivity data, that the first one of the plurality of communication devices is in a field of view of the first LEO satellite.

16. The method of claim 13, wherein the future connectivity data includes a predicted visibility time of a second LEO satellite to the upcoming primary communication device, and the method further comprises transmitting, with the control computer, a command to the upcoming primary communication device to establish a communication link with the second LEO satellite.

17. The method of claim 13, wherein the connectivity data includes at least one selected from the group consisting of a signal strength of the signal between the respective communication device and the first LEO satellite, a quality of service (QoS) of the signal between the respective communication device and the first LEO satellite, an estimated remaining connection time with the first LEO satellite, and obstruction data.

18. The method of claim 13, wherein the satellite line-of-sight obstruction includes a predicted satellite LoS obstruction, further comprising: predicting a path of the first LEO satellite and determine the predicted satellite LoS obstruction based on a predicted path of the first LEO satellite.

19. The method of claim 13, further comprising: in response to determining the satellite LoS obstruction, determining, based on the image data and a predicted path of the first LEO satellite, an unobstructed location, and transmitting a message to the primary communication device, the message including instructions for a user of the primary communication device to move to the unobstructed location.

20. The method of claim 13, further comprising: predicting a path of a second LEO satellite, and in response to determining the satellite LoS obstruction, transmit a command to the primary communication device to establish a communication link with the second LEO satellite.
